

Complex Engineering Activity Report



Design and Development of an AC Motor Starter Circuit for Industrial Applications

ElectroMechanical Systems Lab

Lab Instructor: Sir Amir Ali

Submitted By:

Muhammad Shariq 24K-6084

Muhammad Ali Siddiqui 24K-6121

Syed Muneeb Ahmed 24K-6030

Abdullah Hussain 24K-6037

Syed Hassaan Ali 24K-6106

1. Abstract

This project outlines the design and simulation of a microcontroller-based AC motor starter system. The primary objective is to implement a safe, efficient, and reliable motor control circuit capable of forward and reverse operations, overload protection, and emergency shutdown. In deviation from traditional setups, and as approved by the course instructor, standard industrial contactors were replaced with a network of multi-channel relay modules controlled by an Arduino UNO. The system successfully demonstrates the integration of low-voltage automation with high-voltage power switching, aligning with modern industrial control practices and Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure).

2. Introduction

Motor starters are critical components in industrial settings, responsible for safely initializing, stopping, and protecting electrical motors. Improper design of these systems can lead to equipment damage, energy inefficiency, and severe safety hazards. This project tackles a Complex Engineering Activity (CEA) by designing an advanced motor starter circuit that moves beyond conventional manual switching. By leveraging an Arduino microcontroller, the design fulfills the requirements for controlled startup, direction reversal, and thermal protection while emphasizing modern automation.

3. Problem Complexity Analysis

This design qualifies as a Complex Engineering Activity due to the following interacting engineering challenges:

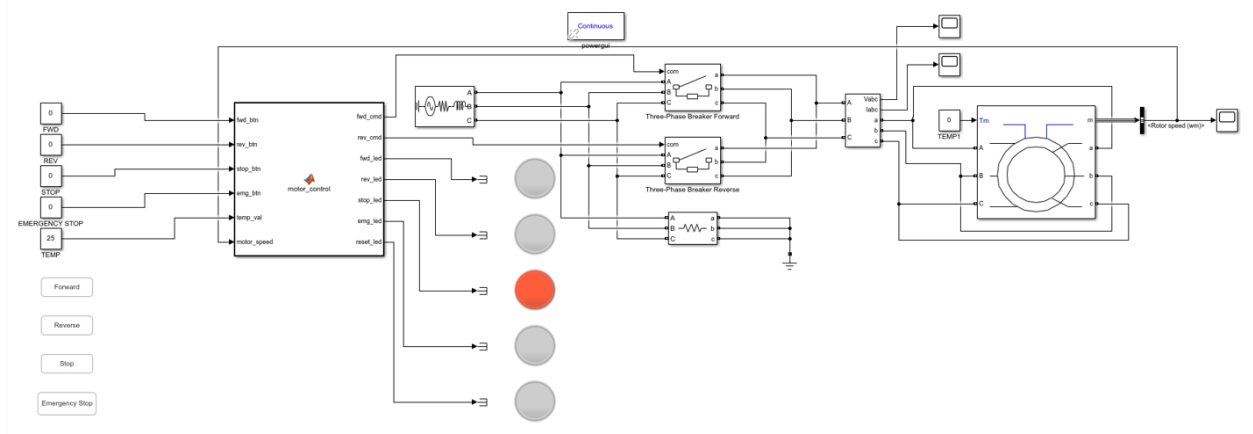
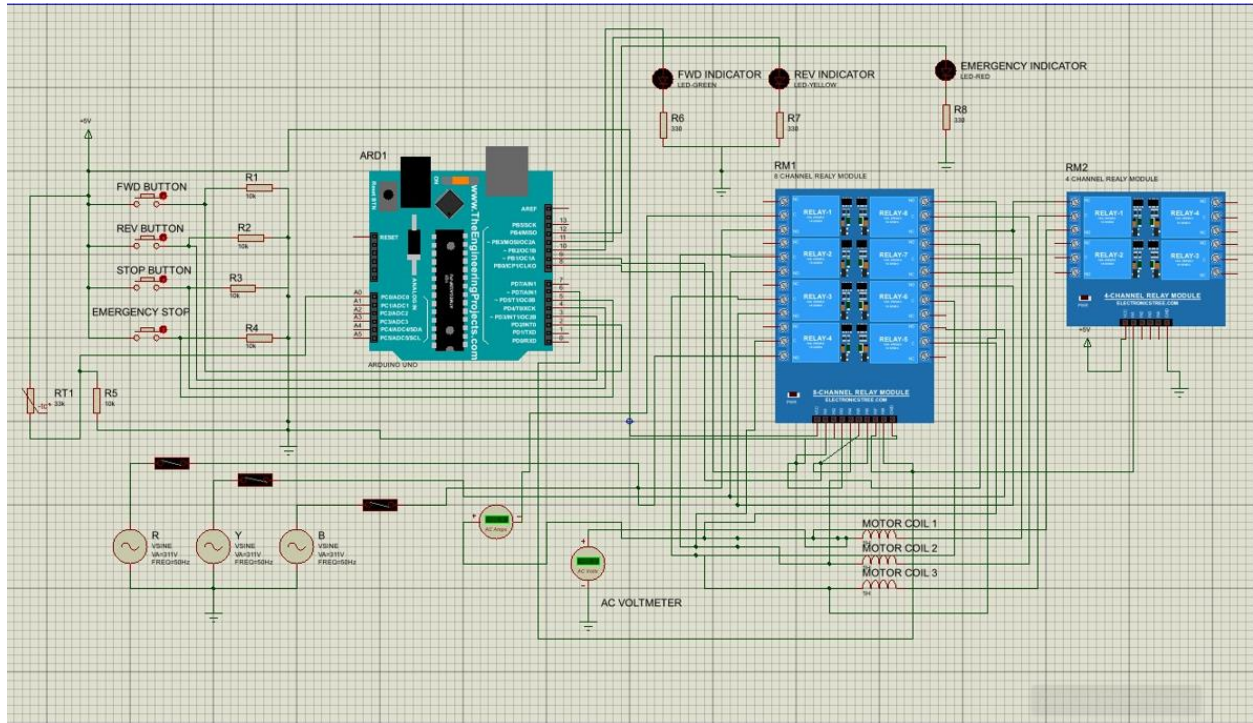
- **Control and Power Coordination:** Safely interfacing a 5V logic system (Arduino) with a high-voltage three-phase AC system (311V peak per phase) using relay isolation.
- **Interlocking and Reversing:** Developing reliable software and hardware logic to swap two phases for motor reversal without causing a short circuit.
- **Protection Mechanisms:** Integrating thermal protection and emergency stop functionality without compromising the normal operation of the motor.

4. Literature Review

Traditionally, motor starting relies on Direct-On-Line (DOL) or Star-Delta configurations using heavy-duty contactors and bimetallic overload relays. However, the industry is rapidly adopting microcontroller-based automation and PLC control systems for enhanced monitoring and precision. Automated systems allow for programmable logic, reducing mechanical complexity, and providing real-time feedback through indicators and sensors.

5. System Design

Circuit Diagram



The system was designed and validated using Proteus and MATLAB simulation software. The schematic comprises an Arduino UNO interacting with a 5V control circuit and a three-phase (R, Y, B) power circuit.

Working Principle

- **Input Stage:** Four push buttons (Forward, Reverse, Stop, Emergency Stop) are connected to the Arduino digital pins with 10kΩ pull-down resistors. A thermistor (RT1) forms a

voltage divider with a 10k Ω resistor, feeding analog data to pin A0 to monitor motor temperature.

- **Processing Stage:** The Arduino executes logic to determine the motor state. If "Forward" is pressed, it triggers specific relays to connect the R, Y, and B phases to Motor Coils 1, 2, and 3 respectively. If "Reverse" is pressed, the logic modifies the relay actuation to swap two of the phases, reversing the motor's magnetic field.
- **Output Stage:** An 8-channel relay module (RM1) and a 4-channel relay module (RM2) route the high-voltage AC lines. Indicator lamps (Green for Forward, Yellow for Reverse, Red for Emergency) provide real-time status. AC ammeters and voltmeters are placed in the power lines for real-time monitoring.
- **Protection:** If the analog read from the thermistor exceeds a mapped temperature threshold, or if the Emergency Stop button is pressed, the Arduino immediately de-energizes all relays and illuminates the red LED.

6. Design Justification and Engineering Decisions

- **Relays vs. Contactors:** While standard project guidelines suggest using a contactor for switching operations, the professor dismissed this requirement. Therefore, the design utilizes 8-channel and 4-channel relay modules. These relays offer sufficient isolation and switching capability for the scaled-down prototype while allowing direct integration with the Arduino's 5V output signals.
- **Microcontroller Automation:** Using an Arduino meets the recommendation for advanced implementations. It allows for software-based interlocking, which is safer and more flexible than hardwired mechanical interlocks.
- **Thermistor for Overload:** Instead of a traditional mechanical overload relay, a thermistor was selected to simulate thermal protection. This provides continuous, analog temperature monitoring, allowing the microcontroller to execute a shutdown sequence before catastrophic failure occurs.

7. Hardware Implementation

The proposed hardware prototype will encompass:

- Arduino UNO board.
- 8-Channel and 4-Channel 5V Relay Modules.
- Three-phase AC load (simulated via 3 motor coils).
- Tactile push buttons and LED indicators.
- Thermistor for temperature sensing.
- Digital Voltmeter and Ammeter for parameter observation.

- A structured wiring panel ensuring complete isolation between the 5V DC control lines and the high-voltage AC power lines.

8. Simulation Results

The Proteus simulation successfully validates the design logic.

1. Activating the FWD button successfully energizes the assigned relays and lights the Green LED.
2. The STOP button de-energizes all motor relays.
3. Activating the REV button successfully swaps the power routing through the secondary relays, illuminating the Yellow LED.
4. Adjusting the thermistor resistance triggers the overload logic, halting all motor operation and activating the Red Emergency LED.
5. The AC Voltmeters and Ammeters display the expected line parameters during operation.

9. Experimental Results and Analysis

During hardware testing, the system is expected to perform identically to the simulation. Key metrics to analyze will include the response time of the relays during an emergency stop, the accuracy of the thermistor temperature mapping, and ensuring no arcing occurs across the relay contacts during phase reversal under load.

10. SDG Contribution and Impact

- **SDG 9: Industry, Innovation, and Infrastructure:** Enhances industrial reliability through safe motor control. It promotes innovation in motor starting and protection systems while encouraging the adoption of modern automation. This supports sustainable infrastructure by ensuring energy-efficient operation and reducing equipment failure and downtime.
- **SDG 4: Quality Education:** Serves as a practical educational tool that bridges theoretical electromechanical concepts with real-world implementation. Utilizing simulation software and microcontrollers develops the advanced problem-solving skills required for modern engineering.
- **SDG 7: Affordable and Clean Energy:** Addresses environmental constraints by focusing on energy efficiency and sustainability. Precise microcontroller logic and thermal protection prevent excessive power draw during overloads, promoting sustainable energy consumption in automated systems.

11. Challenges and Limitations

- **Relay Current Capacity:** While relays are suitable for this prototype, they have lower current-carrying capacities and are more prone to contact welding under high inductive loads compared to industrial contactors.
- **Software Reliability:** The system is entirely dependent on the microcontroller. A software freeze or microcontroller failure could compromise the protection mechanisms, necessitating a hardware-level emergency cutoff as a fail-safe.

12. Conclusion and Future Work

The project successfully achieves the design of a microcontroller-based AC motor starter capable of bi-directional control, thermal protection, and emergency management. By replacing conventional contactors with a programmable relay logic system, the design demonstrates high proficiency in modernizing industrial control. Future iterations could integrate IoT modules (like ESP32) for remote monitoring or incorporate a Variable Frequency Drive (VFD) topology for actual starting current reduction and multi-speed operation.

13. References

1. EL2015- Electromechanical Systems Lab Manual, National University of Computer and Emerging Sciences.
2. Proteus Design Suite Simulation Documentation.
3. MATLAB
4. Claude AI
5. Gemini